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B.Tech/M.Tech (Integrated) DEGREE EXAMINATION, JULY 2025

First & Second Semester

21EES101T - ELECTRICAL AND ELECTRONICS ENGINEERING

(For the candidates admitted during the academic year 2021-2022 to 2024-2025)

Note:

i. **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.

ii. **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 hours **Max. Marks: 75**

PART - A (20 × 1 = 20 Marks)

Answer **all** Questions

		Marks	BL	CO
1.	In a pure resistive circuit, the current _____ the voltage. (A) InPhase (B) Lags (C) Leads (D) Greater than or equal to	1	1	1
2.	R1 = 1Ω, R2 = 3Ω, R3 = 5Ω and R4 = 7Ω connected in series. If the total applied voltage is 20V, the current flowing through the circuit is (A) 2.25 A (B) 1.25 A (C) 12.5 A (D) 125A	1	2	1
3.	For a three-phase star connected system with a line voltage of 440 V, calculate the value of phase voltage. (A) 440V (B) 331.3V (C) 254.03V (D) 230V	1	2	1
4.	The internal resistance for the maximum transfer of power should be (A) Equal to load resistance (B) Greater than load resistance (C) Zero (D) Lesser than load resistance	1	1	1
5.	Which of the following terminal does not belong to BJT? (A) Base (B) Emitter (C) Collector (D) Drain	1	1	2
6.	SMPS stands for (A) Switched Mode Power Supply (B) Switch Maximum Power Supply (C) Silicon Mode Power Supply (D) Switched Mode Power Section	1	1	2
7.	The AND gate output C=1, if the two inputs are (A) A=0, B=0 (B) A=0, B=1 (C) A=1, B=0 (D) A=1, B=1	1	2	2
8.	An SCR is a (A) Four junction device (B) Two junction device (C) Three junction device (D) One Junction device	1	1	2
9.	The brushes in DC machine are made of (A) Rubber (B) Carbon (C) Aluminum (D) Copper	1	1	3
10.	Which of the following is not a part of transformer? (A) Breather (B) Conservator (C) Buchholz relay (D) Commutator	1	1	3
11.	Which of the following motor rotates in incremental step angle? (A) Servo motor (B) DC motor (C) Stepper motor (D) Linear Induction Motor	1	2	3

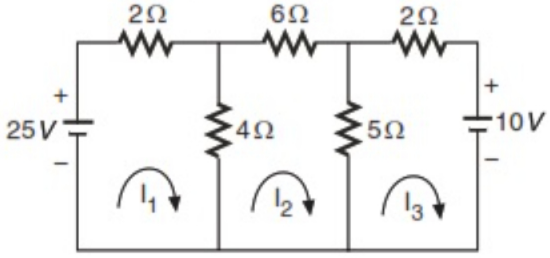
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12.	The stator core of a 3- phase induction motor is laminated in order to reduce the (A) Eddy current loss (B) Hysteresis loss (C) Both eddy current and hysteresis loss (D) Weight of the stator	1	2	3
13.	Moving coil instruments are used to measure (A) DC quantity only (B) AC quantity only (C) Both AC and DC quantity (D) Either AC or DC	1	2	4
14.	____ is the example of an active transducer. (A) Strain gauge (B) Thermistor (C) LVDT (D) Thermocouple	1	1	4
15.	LEDs fabricated from the gallium arsenide emit radiation in the (A) Infrared Region (B) Visible Range (C) Ultraviolet Region (D) Ultrasonic Region	1	1	4
16.	In PMMC instruments, the damping torque is provided by (A) Air friction (B) Magnetic damping (C) Gravity (D) Eddy current damping	1	1	4
17.	Which one of the following is false statement about DC power transmission? (A) The amount of insulation required is more as compared to that for A.C transmission (B) It improves the stability of the system (C) There is no changing current (D) Skin effect increases the effective resistance of the line conductors.	1	1	5
18.	A device that converts optical radiation into electrical energy is (A) Diode (B) BJT (C) MOSFET (D) Solar cell	1	1	5
19.	Which of the following is not a renewable energy recourse? (A) Solar power (B) Wind power (C) Thermal power (D) Tidal power	1	1	5
20.	The most common fuel used in fuel cell is ____ (A) Water (B) Hydrogen (C) Thermal Power (D) Sulphur	1	1	5

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PART - B (5 × 8 = 40 Marks) — Marks | BL | CO

Answer **all** Questions

21.	(a) Determine voltage across the 6Ω resistor using Mesh analysis.  	8	3	1

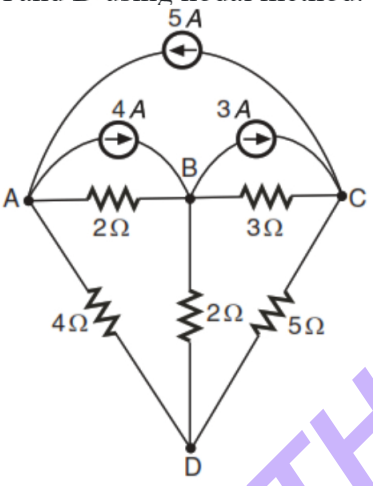
(OR)

(b) In a series circuit containing pure resistance and pure inductance. The current and voltage expressed as $i(t)=5\sin(314t+2\pi/3)$ and $V(t)=20(314t+5\pi/6)$.
(i) What is the impedance of the circuit?
(ii) Calculate the values of resistance, inductance and power factor.

22.	(a) Discuss the input and output characteristics of BJT common base configuration using necessary circuit diagram.	8	3	2
	(OR)			
	(b) Explain the working and characteristics of SCR.			
23.	(a) Describe the construction and working principle of single-phase transformer	8	3	3
	(OR)			
	(b) With the help of block diagram, Explain the operation of chopper fed DC drive.			
24.	(a) Analyse the working principle of repulsion type moving iron instrument with neat diagram. Mention its Advantages.	8	4	4
	(OR)			
	(b) Describe the operation of LVDT with its characteristics curve.			
25.	(a) Write notes on EV charging station with the required components.	8	3	5
	(OR)			
	(b) What is the necessity of Earthing? Explain the different types of earthing methods.			

PART - C (1 × 15 = 15 Marks) **Marks BL CO**

Answer **any 1** Questions

26.	In the circuit given in the following figure, Compute the potential difference between A and D using nodal method. 	15	4	1
27.	(i) Realize the logic circuit for the following Boolean expression (7 Marks) F= A B C+A B C' +A' B C' (ii) Simplify the following function using k-map and implement the simplified expression using logic gates (8 Marks) Y(A ,B, C,D) = Σm (0,1,2,4,5,7,8,9,10,12,13)	15	4	2

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